

Appointments

8.

$$X \sim N(12, 16)$$

$$4 \quad P(X \leq c) = 2P(X > c) \text{ find } c.$$

$$P(X \leq c) = 2[1 - P(X \leq c)]$$

$$\Rightarrow 3P(X \leq c) = 2$$

$$\Rightarrow P(X \leq c) = \frac{2}{3} = 0.666$$

$Z = \frac{X - \mu}{\sigma}$
standardization

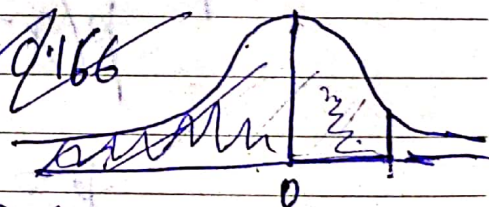
$$P(Z \leq \frac{c - 12}{4}) = 0.666$$

Phones to make

$$\Rightarrow P(Z \leq \frac{c - 12}{4}) = 0.666$$

$$\Rightarrow \Phi\left(\frac{c - 12}{4}\right) = 0.666$$

$$\Rightarrow \frac{c - 12}{4} = 0.43$$



$$\Rightarrow c = 12 + 4 \times 0.43$$

Remarks

$$\Phi\left(\frac{c - 12}{4}\right) - \Phi(0) = 0.166$$

$$\Rightarrow \Phi\left(\frac{c - 12}{4}\right) = 0.166$$

$$\Rightarrow \frac{c - 12}{4} = 0.43$$

$$\Rightarrow c = 12 + 4 \times 0.43$$

$$= 12 + 1.72$$

$$= 13.72$$

DECEMBER

S S M T W T F S S S M T W T F S S S M T W T F S S M
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31

Appointments

(1)

The mean yield for one acre plot is 662 kilos with 1.132 kilos. Assuming normal distribution, how many one acre plots in a batch of 1,000 plot would you expect to have yield over 700 kilos.

$$X \sim N(\mu, \sigma), \mu = 662, \sigma = 32$$

Phones to make

$$\therefore P(X > 700) = P(Z > 1.19)$$

$$Z = \frac{X - 662}{32}$$

$$= .5 - P(0 \leq Z \leq 1.19)$$

$$= .5 - .3830$$

$$= .117$$

$$\therefore \text{expected no.} = 1,000 \times .117 = 117$$

Remarks

$$\begin{aligned} P(A' \cap B') &= 1 - P(A) - P(B) + P(A \cap B) \\ &= P(A') - P(B) P(A') \\ &= P(A') P(B') \end{aligned}$$

JUNE

F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	

Log Normal Distribution

A positive, ^{continuous} r.v. X is said to have a log-normal distribution if $\log_e X$ is normally distributed; i.e.

$$\text{let } Y = \log_e X \sim N(\mu, \sigma^2)$$

$$\text{pdf } f(x) = \frac{1}{x} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(\ln x - \mu)^2}{2\sigma^2}}$$

Phones to make

Moment generating function does not exist.

characteristic function exists

~~pdf~~

Remarks

$$\text{mean } \mu_Y = \ln\left(\frac{\mu_X^2}{\sqrt{\mu_X^2 + \sigma_X^2}}\right)$$

$$\text{sd } \sigma_Y = \ln\left(1 + \frac{\sigma_X^2}{\mu_X^2}\right)$$

$$\therefore \text{cdf} = \frac{1}{\sigma\sqrt{2\pi}} \int \log u e^{-\frac{(Y-\mu)^2}{2\sigma^2}} dy$$

FEBRUARY

T F S S M T W T F S S M T W T F S S M T W T F S S M T W
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28

7th. Order raw moment

$$\mu'_r = E(e^{rx}) = e^{(\mu r + \frac{1}{2} r^2 \sigma^2)}$$

if in particular we assume

$$\mu = \log \alpha, \quad \sigma > 0$$

$$\text{i.e. } \log x \sim N(\log \alpha, \sigma^2)$$

$$\text{then } \mu'_r = \alpha^r e^{\frac{1}{2} r^2 \sigma^2}$$

$$\text{mean } \mu'_1 = \alpha e^{\frac{1}{2} \sigma^2}, \quad \mu'_2 = \alpha^2 e^{\sigma^2}$$

$$\therefore \text{Variance} = \alpha^2 e^{\sigma^2} (e^{\sigma^2} - 1)$$

Properties.

① If $x_1, x_2, \dots, x_n$ is a set of independent, identically distributed r.v. such that mean of each $\log x_i$ is μ and variance σ^2 , then the product $x_1, x_2, \dots, x_n$ is asymptotically distributed according to logarithmic normal distribution and with mean μ and variance σ^2 .

T F S S M T W T F S S M T W T F S S M T W T F S

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31

MARCH

Appointments

2. Log-normal distribution
is positively skewed and
mean > median > mode.
mean = $e^{\mu + \sigma^2/2}$, median = e^{μ}
mode = $e^{\mu - \sigma^2}$

Phones to make

~~$y = \ln X$~~ $y = \ln X \sim N(\mu, \sigma^2)$

Remarks

MARCH

T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31